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10 October 1975
SUPERSEDING
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MILITARY SPECIFICATION

TRACTOR, FULL-TRACKED, LOW-SPEED; TRACTOR WHEELED, AGRICULTURAL; AND TRACTOR WHEELED, INDUSTRIAL: AND THEIR ATTACHMENTS, PACKAGING OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for materials, methods and procedures for the preservation, packing, and marking of tractors, full-tracked, low-speed; wheeled, agricultural; and wheeled, industrial, and their attachments.

1.2 Classification. Preservation, and packing shall be one of the following types, as specified (see 6.2):

Type I - Boxed.
Type II - Unboxed (mobile).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
QQ-S-781	- Strapping, Steel, and Seals.
TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.

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TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-1055	- Barrier Material, Waterproofed, Flexible.
PPP-C-650	- Crates, Wood, Open and Covered.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.

Military

MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation-Packaging, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-R-196	- Repair Parts for Internal Combustion Engines, Packaging of.
MIL-L-2105	- Lubricating Oil, Gear, Multipurpose.
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-C-5501	- Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-B-13239	- Barrier Material, Waterproofed, Flexible, All Temperatures.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-E-16298	- Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-P-46046	- Preservative Fluid, Automotive Brake System and Components.
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.

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STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

Federal

FED. STD. No. 356	- Commercial Packaging of Supplies and Equipment.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein and in applicable specifications. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification. Materials shall be free of defects.

3.2 Preproduction pack. The supplier shall furnish a preproduction pack for examination and testing within the time frame specified (see 6.2) to prove prior to starting production packaging that the applied preservation, packing, and marking comply with the requirements of this specification. Examination and test shall be those specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Repainting. Painted surfaces of the tractors and attachments on which the paint has been damaged shall be cleaned and repainted in accordance with the painting requirements of the end item specification (see 6.4).

3.4 Preservation. Preservation shall be Level A, B or commercial, as specified (see 6.2).

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3.4.1 Level A.

3.4.1.1 Disassembly. Disassembly shall be confined to the minimum necessary to safeguard parts vulnerable to damage, pilferage and loss, except as required by the transportability requirements of the end item specification. The extent of further disassembly shall be confined to the removal of protruding parts that would otherwise increase cubage; however, disassembly shall be limited to those parts that can be removed and reinstalled without special tools. Removed bolts, nuts, screws, pins, and washers shall be reinstalled in mating parts and secured to prevent their loss. When it is not practical to reinstall the hardware, the items shall be preserved in accordance with MIL-P-116, Method IC-1, and the preserved items secured to the component from which they had been removed. Keys removed shall be secured in the keyway of the primary component or shaft.

3.4.1.2 Matchmarking. Parts removed and mating parts on the tractors and attachments shall be matchmarked when necessary to facilitate reassembly. Large parts shall be matchmarked identically by stenciled letters or numerals with lusterless white enamel, overcoated with varnish conforming to TT-V-121, or identically matchmarked with paint conforming to MIL-P-13983. Small parts removed and mating parts on the tractors shall be matchmarked and identified with tags as specified in 3.4.1.3.

3.4.1.3 Tags. Tags used for matchmarking and instructions shall conform to UU-T-81, Type A, or UU-T-81, Type B, Class 2. Marking of the tags shall be with waterproof ink.

3.4.1.4 Cleaning, drying, and preservatives. Metal surfaces to which a preservative is to be applied shall be cleaned and dried in accordance with MIL-P-116. Preservatives identified herein by "P" numbers shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

3.4.1.5 Unprotected metal surfaces. Unpainted exterior metal surfaces of tractors and attachments, except machined surfaces, requiring the application of a contact preservative in accordance with MIL-P-116, such as tie rods, adjusting rods, clevises, bolts, nuts, coil springs, leaf springs, cable sheaves pins, exposed linkages, grease fittings, drawbars, and any other surface including those surfaces exposed by disassembly, shall be coated with Type P-1 preservative. Exterior machined surfaces, where removal is required to place the tractor in operation, shall be coated with Type P-6 or P-11 preservative. Other machined surfaces, where removal is not required, shall be coated with Type P-1 preservative. Surfaces coated with Type P-6 or P-11 preservative shall be wrapped or covered with greaseproof barrier material conforming to MIL-B-121, Type I, Grade A, Class 2, and secured in place with tape conforming to PPP-T-60, Type IV, Class 1.

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3.4.1.6 Engines. Engines and engine accessories shall be preserved in accordance with MIL-E-10062, Level A, Type II, Method I, for boxed tractors, and MIL-E-10062, Level A, Type I, Method I, for unboxed (mobile) tractors.

3.4.1.7 Winterization components.

3.4.1.7.1 Pumps, burners, torches, and piping. Interior surfaces of pumps, burners, torches, and piping requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with Type P-10, Type I, Grade 10 preservative and the excess preservative allowed to drain. The threaded surfaces of fillcaps and drainplugs shall be coated with the preservative specified herein and the caps and plugs reinstalled. Openings in burners not closed with caps and plugs shall be closed with tape conforming to PPP-T-60, Type IV, Class 1. Blowtorches and blowtorch-type burners shall each be placed in a fiberboard box conforming to PPP-B-636, W5c or W6c. The box shall be closed in accordance with Method V of the appendix to the box specification.

3.4.1.7.2 Fuel tanks. The fuel tanks for heater fuel, if not equipped with baffles, shall be fogged with Type P-10, Type I, Grade 10 preservative, and the excess preservative drained. Fuel tanks with baffles shall be completely filled with the preservative and drained. The drainplug and fillcap threads shall be coated with the preservative, and the drainplug and fillcap shall be reinstalled. The preservative may be reused for processing other tanks but shall be discarded when more than 10 percent of the resultant mixture is fuel (see 6.5).

3.4.1.7.3 Electric motors. Electric motors shall be preserved in accordance with MIL-E-16298, Level A, using the alternate method required for equipment assembled to machines.

3.4.1.8 Drive shafts and universal joints. Unprotected surfaces of drive shafts and universal joints shall be coated with Type P-1 preservative. Dust boots shall be reinstalled after the preservative has dried.

3.4.1.9 Gears.

3.4.1.9.1 Exposed gears. Unprotected surfaces of exposed gears shall be coated with Type P-1 preservative or with primer conforming to TT-P-664.

3.4.1.9.2 Enclosed gears. Gears operating on lubricating oil (SAE 10, 30 or 50) shall have the housing filled to the operating level with preservative conforming to Type P-10, Type I, grade as applicable. The gear housings shall be tagged to indicate: "The preservative oil contained in this gear housing is good for operation. Do not drain until the first required

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lubricant change." Gears operating on gear lubricant (SAE 80 or 90) shall be filled to the operating level with the applicable grade of lubricant conforming to MIL-L-2105. The gear housings shall be tagged to indicate: "The lubricant contained in this gear housing is good for operation. Do not drain until the first required lubricant change." Gears not operating on lubricating oil (SAE 10, 30 or 50) or gear lubricant (SAE 80 or 90) shall have the housing filled to the operating level with the approved lubricant required for operation. The gear housings shall be tagged to indicate: "This gear housing is filled to the operating level with the applicable type lubricant required for operation. Do not drain until the first required lubricant change." Tags shall be as specified in 3.4.1.3.

3.4.1.10 Drive chains.

3.4.1.10.1 Exposed drive chains. Exposed drive chains shall be coated with enough Type P-3 or P-9 preservative to insure penetration of the preservative to the inner surfaces of the rollers, pins and bushings. The entire chain shall be coated with Type P-1 preservative after the excess preservative has drained.

3.4.1.10.2 Enclosed drive chains. Enclosed drive chains shall be preserved as specified in 3.4.1.9.2 for enclosed gears.

3.4.1.11 Clutches. Clutches shall not be operated after being preserved as specified herein.

3.4.1.11.1 Disc type (dry type). With the cover plates removed and the clutch engaged, all accessible metal interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664. When specified (see 6.2), the clutch shall be disassembled sufficiently to permit the application of a continuous thin film (maximum 0.75 ml) of the same primer to friction surfaces of the pressure plates. Clutch control mechanisms not enclosed shall be coated with Type P-1 preservative. After being sprayed, spring-loaded-type clutches shall have the clutch pedal secured in a partially disengaged position to eliminate contact between the disc facings and the pressure plates. Snap-over-center- and toggle-in-type clutches shall be completely disengaged. Cover plates shall be reinstalled.

3.4.1.11.2 Band and shoe type. Clutch drum facings and control mechanisms enclosed in the housings shall be coated with primer conforming to TT-P-664. Clutch control mechanisms not enclosed in housings shall be coated with Type P-1 preservative.

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3.4.1.11.3 Cone type. The unlined cones and control mechanisms enclosed within housings shall be coated with primer conforming to TT-P-664. Control mechanisms not enclosed in housings shall be coated with Type P-1 preservative. The clutch shall be disengaged to prevent the linings from adhering to the unlined cones.

3.4.1.11.4 Jaw type. Clutch jaws, shifter yokes, and machined surfaces of shafts shall be coated with Type P-1 preservative. Control mechanisms enclosed within housings shall be coated with Type P-10, Type I, Grade 10 preservative. Control mechanisms not enclosed within housings shall be coated with Type P-1 preservative.

3.4.1.11.5 Flywheel clutch, oil type (running in oil). The clutch shall be preserved at the same time as the engine preservative run-in. The housing shall be filled to the operating level with preservative lubricating oil conforming to Type P-10, Type I, grade as applicable. The fill plug threads shall be coated with the preservative lubricating oil and the fill plug reinstalled. When the engine reaches its normal operating temperature, the clutch shall be operated to insure coating of all parts and surfaces with the preservative. Upon completion of preservation, all openings into the clutch housings shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1. A tag shall be attached to the clutch control indicating: "This clutch housing is filled to the operating level with preservative lubricating oil, applicable grade, and is suitable for operation until the first required lubricant change. Remove tape from openings prior to operation of the clutch." Tags shall be as specified in 3.4.1.3.

3.4.1.11.6 Flywheel clutch, oil type (lubricated by engine lubricating system). The clutch shall be processed at the same time as the engine preservative run-in. When the engine reaches its normal operating temperature, the clutch shall be operated to insure coating of all parts and surfaces with the lubricant. Upon completion of processing, all openings into the clutch housing shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1. A tag shall be attached to the clutch control indicating "Remove tape from openings prior to operation of the clutch." Tags shall be as specified in 3.4.1.3.

3.4.1.11.7 Steering clutches.

3.4.1.11.7.1 Dry type. Dry disc-type steering clutches shall be preserved as specified in 3.4.1.11.1. Other dry-type steering clutches shall be preserved as specified herein for the applicable type of dry clutch.

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3.4.1.11.7.2 Oil type. The clutch housing shall be filled to the operating level with preservative lubricating oil conforming to Type P-10, Type I, grade as applicable. The fill plug threads shall be coated with the preservative lubricating oil and the fill plug shall be reinstalled. The clutch shall be operated to insure coating all parts and surfaces with the preservative. Upon completion of preservation, all openings into the clutch housings shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1. A tag shall be attached to the clutch control indicating: "The housing is filled to the operating level with preservative lubricating oil of applicable grade, and is suitable for operation until the first required lubricant change. Remove tape from openings prior to operation of the clutch." Tags shall be as specified in 3.4.1.3.

3.4.1.11.7.3 Hydraulic boosters (having a separate oil supply). Hydraulic boosters having a separate oil supply reservoir shall be preserved at the same time as the steering clutch. The oil reservoir shall be filled to the operating level with preservative lubricating oil conforming to Type P-10, Type I, grade as applicable. The fill plug threads shall be coated with the preservative lubricating oil and the fill plug reinstalled. The booster shall be operated to insure coating all parts and surfaces with the preservative. Upon completion of preservation, all openings into the system shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1. A tag shall be attached to the clutch control indicating: "The hydraulic booster is filled to the operating level with the applicable grade of preservative lubricating oil and is suitable for operation until the first required lubricant change. Remove tape from openings into the system prior to operating the clutch." Tags shall be as specified in 3.4.1.3.

3.4.1.11.8 Clutch housings. For Type I boxed tractors, drainplugs for dry-type clutches shall be removed from clutch housings. The plugs shall be identified by tags specified in 3.4.1.3. The threads of the plugs and the housing threads shall be coated with preservative lubricating oil conforming to Type P-10, Type I, Grade 10. The plugs shall be secured in a protected and conspicuous location. A tag specified herein shall be secured to the clutch control indicating: "Reinstall drainplug in clutch housing prior to operation of the clutch." For Type II unboxed (mobile) tractors, openings into clutch housings shall be covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2, and secured in place with tape conforming to PPP-T-60, Type IV, Class 1. A tag as specified herein shall be attached to the clutch control indicating: "Remove covers from housing openings prior to operation of the clutch."

3.4.1.12 Brakes.

3.4.1.12.1 General. Exterior, unpainted, untreated, and threaded surfaces of cables, clevises and linkages of service and parking brakes requiring the

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application of a contact preservative in accordance with MIL-P-116, shall be coated with Type P-1 preservative. When specified (see 6.2), metal surfaces enclosed within brakedrums (such as adjusting wedges, pins, eccentrics, cam levers, linkages, anchors, retracting springs) and the braking surface of the brakedrums shall be sprayed with a continuous film of primer conforming to TT-P-664. The film shall be a minimum of 0.50 mil to a maximum of 0.75 mil thick as determined by elcometer. Primer material containing lumps or skin that will not return to a homogenous mixture upon stirring shall not be used. The film shall be thoroughly dry prior to reinstalling the brakedrums. A warning tag shall be securely attached to the brake control indicating: "Brakedrums and components preserved. Do not apply brakes when tractor is being moved. Towing shall be accomplished by use of a rigid towbar or similar arrangement until this tractor is issued to the user." Tags shall be as specified in 3.4.1.3.

3.4.1.12.2 Hydraulic brake systems.

3.4.1.12.2.1 Systems operating on nonpetroleum base fluid. The hydraulic brake system shall be filled to the operating level with preservative fluid conforming to MIL-P-46046 and the brakes actuated to insure coating all interior parts and surfaces with the preservative fluid. A tag shall be attached to the brake control indicating: "The brake system of this tractor is filled to the operating level with preservation fluid and is good for indefinite storage and limited operation in ambient temperatures not lower than minus 10° F. Prior to regular operation in ambient temperatures lower than minus 10° F, the brake system shall be drained and refilled to the operating level with the nonpetroleum base fluid recommended for operation." Tags shall be as specified in 3.4.1.3.

3.4.1.12.2.2 Systems operating on petroleum base fluid. The hydraulic systems of brakes and steering systems that operate on petroleum base hydraulic fluids shall be filled to the operating level with the fluid required for operation and the brakes and steering systems shall be actuated to insure coating of all interior surfaces with the fluid. A tag shall be attached to the system controls indicating: "This system is filled with the approved operating fluid. Do not drain." Tags shall be as specified in 3.4.1.3.

3.4.1.12.3 Air-actuated brake systems. The air-actuated brake systems shall be drained. Interior surfaces of the air-supply tanks shall be fogged with Type P-10, Type I, Grade 10 preservative. Excess preservative lubricating oil shall be drained. Threaded openings and the threads of drainplugs shall be coated with preservative lubricating oil specified herein. The drainplugs shall be reinstalled. Air-line filters shall be drained and closed. The exhaust ports of relay emergency, quick-release, and relay valves not equipped with exhaust check valves shall be closed by inserting

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plugs conforming to MIL-C-5501 or by sealing the ports with tape conforming to PPP-T-60, Type IV, Class 1. Tags shall be attached to the valves and service lines indicating: "The plugs or tape must be removed from the exhaust ports and valves before this tractor is placed in operation." Tags shall be as specified in 3.4.1.3.

3.4.1.12.4 Air-hydraulic brakes. Air-hydraulic brakes shall be preserved in accordance with 3.4.1.12.1, 3.4.1.12.2 and 3.4.1.12.3. To preserve the interior surfaces, the plugs shall be removed from housings and connectors and the preservative lubricating oil shall be sprayed through each opening. The plugs and hose shall be reinstalled, and the brake pedal shall be operated through two cycles. The plugs and hose shall again be removed, and the interior surfaces shall again be sprayed as specified herein. The plugs and hose shall again be reinstalled.

3.4.1.13 Hydraulic systems (except hydraulic brakes or steering systems). The hydraulic fluid supply tanks shall be filled to the operating level with hydraulic fluid required for operation. The pistons shall be retracted as far as practicable into the cylinders and secured. When the pistons cannot be fully retracted, the exposed portions of the piston rods (ramshafts) shall be coated with Type P-6 or P-11 preservative and the coated surfaces wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2, extending the wraps approximately 2 inches on the ram cylinders. The wraps shall be secured in place with tape conforming to PPP-T-60, Type IV, Class 1. When the pistons can be fully retracted, any remaining uncoated surfaces of the piston rods shall be coated with Type P-1 preservative, with no wrapping required. The hydraulic control valves shall be secured in the neutral position and preserved as specified herein for piston rods. Hoses shall not be disconnected. Any unpainted exterior metal surfaces of the system requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. A tag shall be attached to the control lever indicating: "The hydraulic supply tank is filled to the operating level with fluid required for operation. Do not drain." Tags shall be as specified in 3.4.1.3.

3.4.1.14 Automatic transmissions, fluid couplings, and torque conversion transmissions.

3.4.1.14.1 Automatic transmissions and fluid couplings. The housings shall be filled to the operating level with fluid or lubricant recommended for operation. The transmissions and fluid couplings shall be operated at sufficient engine speed to insure coating of all interior surfaces with the fluid or lubricant. Openings into the housings shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1. A tag shall be attached to the

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control lever indicating: "The housing is filled to the operating level with the approved fluid or lubricant and is good for operation of this tractor. Do not drain. Do not operate tractor until the tape has been removed from the openings." Tags shall be as specified in 3.4.1.3.

3.4.1.14.2 Torque conversion transmissions.

3.4.1.14.2.1 Lubricating oil operated. The converter shall be preserved at the same time as the engine. At the beginning of engine preservation, the converter housing shall be filled to the operating level with the approved operating oil. The converter shall be operated a sufficient length of time during fuel system preservation to insure coating of all interior surfaces with the operating oil. Openings into the system shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1. A tag shall be attached to the converter control indicating: "The converter housing is filled to the operating level with the approved oil required for operation. Do not drain. Remove tape from housing openings prior to operation of this tractor." Tags shall be as specified in 3.4.1.3.

3.4.1.14.2.2 Diesel fuel operated. The converter shall be preserved at the same time as the engine. At the beginning of engine preservation, the system shall be drained of diesel fuel. The converter system shall be filled to the operating level with Type P-9 preservative and the converter operated a sufficient length of time during fuel system preservation to insure coating of all interior parts with the Type P-9 preservative. The Type P-9 preservative shall remain in the system. Openings into the system shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1. A tag shall be attached to the converter control indicating: "The converter system is filled to the operating level with Type P-9 preservative. Do not operate this tractor until the Type P-9 has been drained and the system has been filled to the operating level with the approved operating fuel. Remove tape from openings into housing." Tags shall be as specified in 3.4.1.3.

3.4.1.15 Wheels, axles and bearings. Wheels and hubs, if removed from the axles or spindles for Type I boxed tractors, shall have the interior bearing surfaces of the axles and spindles coated with Type P-6 or P-11 preservative. Coated surfaces of the axles or spindles shall be wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2. The wrap or cover shall be secured in place with tape conforming to PPP-T-60, Type IV, Class 1. Bearings removed shall be coated with Type P-6 or P-11 preservative and shall be individually preserved in accordance with MIL-P-116, Method IC-1. Openings of the hubs shall be covered with barrier material as specified herein and the barrier material secured in place with the tape specified herein.

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3.4.1.16 Rubber tires. Tires mounted on the wheels and rims (road tires) shall be inflated to 10 pounds above the pressure value recommended for maximum load. Tires mounted on wheels and rims (spare tires) shall be inflated to two-thirds of the specified operating pressure.

3.4.1.17 Crawlers. Surfaces of unpainted metal crawler treads, tracks, sprockets, bogie wheels, rolls, pins, linkages, grouser tips, roller surfaces, and link rails that are readily accessible shall be coated with Type P-1 preservative. However, absence of preservative on these components will not impair their usefulness.

3.4.1.18 Instrument panels. For Type II unboxed (mobile) tractors, all openings into equipment mounted on instrument panels shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1. In addition, for Type II unboxed (mobile) tractors that are not equipped with closed cabs or the panel is not equipped with a vandal guard, the dial glasses of instruments mounted on the panels shall be individually covered or the entire panel may be covered when practicable, with a fitted piece of hardboard or 1/4-inch plywood secured in place with tape as specified herein.

3.4.1.19 Trouble lights, lamps, reflectors, and rear-vision mirrors. Trouble lights, removed lamps, removed reflectors, and removed rear-vision mirrors shall be individually placed in boxes conforming to PPP-B-636, W5c or W6c, and the boxes sealed in accordance with the appendix to the box specification, Method V. Lamps, reflectors and mirrors not removed from Type II unboxed (mobile) tractors shall be covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2. The barrier material shall be secured in place with tape conforming to PPP-T-60, Type IV, Class 1.

3.4.1.20 Gearshifts and control levers. Unpainted metal surfaces of handles and levers shall be coated with Type P-1 preservative. For Type II unboxed (mobile) tractors, composition and rubber handgrips (if not enclosed within a full cab) shall be removed and preserved in accordance with MIL-P-116, Method IC-1, and placed in the tractor toolbox, or shall be consolidated as specified in 3.4.1.33. The threaded surfaces, exposed by removal of handgrips, shall be coated with Type P-1 preservative. When the composition or rubber handgrips are left in place for any reason the grips shall be wrapped with barrier material conforming to PPP-B-1055, Class E-2 and the wrap secured in place with tape conforming to PPP-T-60, Type IV, Class 1.

3.4.1.21 Drivebelts and pulleys. Drivebelts shall be removed or released from tension. Unpainted pulley grooves shall be coated with primer conforming to TT-P-664. Belts removed shall be preserved in accordance with MIL-P-116, Method IC-1. A tag shall be attached in a conspicuous location within the operator's compartment indicating: "Drivebelts have been released from tension or removed. Check installation prior to operation." Tags shall be as specified in 3.4.1.3.

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3.4.1.22 Seatbacks, armrests and cushions. Type II unboxed (mobile) tractors not equipped with closed cabs shall have the seatbacks, armrests and cushions that are removable, removed and individually wrapped with barrier material conforming to MIL-B-13239, Type B-2, Grade A, or wrapped with black polyethylene conforming to L-P-378 (6 mil). The barrier material or plastic film shall be secured with tape conforming to PPP-T-60, Type IV, Class 1. The removed items shall be consolidated as specified in 3.4.1.32.2. Seatbacks, armrests and seat cushions that are not removable for Type II unboxed (mobile) tractors not equipped with fully closed cabs, shall be wrapped or covered in place with the barrier material or plastic film specified herein and the wraps secured with tape as specified herein. A 2-inch opening for drainage shall be made in the barrier material or plastic film at the underside of the seatbacks, armrests and cushions.

3.4.1.23 Windshield wiper arms and blades. For Type II unboxed (mobile) tractors, windshield wiper arms and blades shall be removed and preserved in accordance with MIL-P-116, Method IC-1. The preserved wiper arms and blades shall be placed in the toolbox or shall be consolidated as specified in 3.4.1.33. The exposed wiper shafts shall be coated with Type P-1 preservative.

3.4.1.24 Locks, keys and key openings. Interiors of locks shall be coated with powdered graphite. Openings into exterior locks for Type II unboxed (mobile) tractors shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1. Switch keys, cab door keys, and padlocks with keys shall be preserved in accordance with MIL-P-116, Method IC-1. The preserved keys and padlocks with keys shall be placed in the toolbox. A tag shall be attached in a conspicuous location in the operator's compartment indicating: "Keys for this tractor are preserved and placed in the toolbox." Tags shall be as specified in 3.4.1.3.

3.4.1.25 Hinges and fasteners. Hinges and metal fasteners shall be coated with Type P-10, Type I, Grade 10 preservative.

3.4.1.26 Engine hoods. Engine hoods shall be closed and secured. For Type II unboxed (mobile) tractors, vents, louvers, and other openings into the engine compartment shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1, or the openings shall be covered with barrier material conforming to PPP-B-1055, Class E-2, and the cover secured in place with tape as specified herein.

3.4.1.27 Radiator fronts. When specified (see 6.2), for Type II unboxed (mobile) tractors, the radiator fronts shall be covered with barrier material conforming to PPP-B-1055, Class E-2, and the barrier material secured in place with tape conforming to PPP-T-60, Type IV, Class 1.

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3.4.1.28 Attachments.

3.4.1.28.1 Power takeoff and power control units. Power takeoff and power control units shall have the components preserved as specified herein for parts and assemblies of similar design and having similar characteristics.

3.4.1.28.2 Winches. Components of winches shall be preserved as specified herein for components of similar design and having similar characteristics.

3.4.1.28.3 Wire rope. Wire rope (except galvanized) shall be coated with Type P-1 preservative applied by drawing the wire rope through a trough filled with preservative over an arrangement of three pulleys or by any other convenient manner so that the rope will be completely immersed in the preservative. Excess preservative shall be removed by a swab around a cable guide in the end of the trough. The wire rope shall be wound on the winch drums. For Type II unboxed (mobile) tractors, the wire rope shall be covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2. The barrier material shall extend around the rope on the drum and be of sufficient width to be taped to the flanges of the drum with tape conforming to PPP-T-60, Type IV, Class 1. A 2-inch opening for drainage shall be made in the barrier material at the underside of the drum.

3.4.1.28.4 Scarifiers. Scarifiers and components shall be preserved as specified herein for parts and assemblies having similar characteristics. Scarifiers mounted on tractors shall be positioned and secured in the transport position.

3.4.1.28.5 Other attachments. Components of other attachments such as dozers, bull clams, loaders, pipe layers, angle fillers, mowers, sweepers, backhoes, cranes, and any other attachments shall be preserved as specified herein for components having similar characteristics.

3.4.1.29 Fire extinguishers. Unpainted exterior surfaces of fire extinguishers requiring the application of a contact preservative in accordance with MIL-P-116 shall be coated with Type P-1 preservative. For Type I boxed tractors, the fire extinguishers shall be secured in the mounting brackets. For Type II unboxed (mobile) tractors, each fire extinguisher shall be placed in a box conforming to PPP-B-636, W5c, or W6c. The boxes shall be waterproofed sealed in accordance with the appendix to the box specification, Method V.

3.4.1.30 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, Level A.

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3.4.1.31 Technical publications. Technical publications for each tractor shall be preserved together in accordance with MIL-P-116, Method IC-1.

3.4.1.32 Repair parts. Unless otherwise specified herein, the preservative application criteria and applicable method(s) of preservation of MIL-P-116 shall be used to preserve the repair parts. When specified, repair parts shall be preserved in accordance with MIL-R-196, Level A; however, when parts are not specifically covered in MIL-R-196, an applicable submethod of preservation of MIL-P-116 shall be used.

3.4.1.33 Consolidation.

3.4.1.33.1 For Type I boxed tractors. The repair parts, tools, technical publications, and all other loose components (except wheels and attachments) requiring the protection of a container shall be consolidated. The toolbox shall be utilized as a container and the lid shall be closed and secured. The remainder of the components that will not fit in the toolbox shall be consolidated in close-fitting boxes conforming to PPP-B-601, Domestic Type, style optional, Grade B, or wood boxes conforming to PPP-B-621, Class 1, style optional. The number, size and weight of the boxes shall be determined by the available space for packing in the crate with the tractor in a manner to conserve cube.

3.4.1.33.2 For Type II unboxed (mobile) tractors. Consolidation shall be as specified in 3.4.1.33.1, except the boxes shall be Overseas Type, and the lid of the toolbox shall be secured to prevent pilferage. Strapping of the boxes shall be with strapping conforming to QQ-S-781, Class 1, Type I or IV, size as required. Strapping shall be Finish B unless otherwise specified (see 6.2). When specified, strapping shall be Finish A. To the maximum extent possible, number, size and weight of the boxes shall be determined by the available space and convenience for securing to the tractor in a manner that will not increase cubage or interfere with towing or lifting the tractor with slings. Otherwise, the boxes shall be shipped separately from but at the same time as the tractor.

3.4.1.33.3 For attachments. Components removed from attachments shall be consolidated as specified in 3.3.1.32.1 for boxed attachments and as specified in 3.4.1.33.2 for unboxed attachments.

3.4.2 Level B. Unless otherwise specified (see 6.2 and 6.6), Level B preservation shall be as follows.

3.4.2.1 Disassembly and matchmarking. Disassembly shall be as specified in 3.4.1.1 for Level A. Matchmarking shall be as specified in 3.4.1.2 for Level A, except tags shall be as specified in 3.4.2.2.

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3.4.2.2 Tags. Tags for matchmarking and instructions shall be as specified in 3.4.1.3 for Level A, except marking ink need not be waterproof.

3.4.2.3 Preservatives. Preservatives and their application shall be as specified in 3.4.1.4 and 3.4.1.5 for Level A.

3.4.2.4 Engines. The fuel systems, combustion chambers, valves, coolant systems, and batteries shall be preserved in accordance with MIL-E-10062, Level A, Type I, Method I for unboxed (mobile) tractors, and MIL-E-10062, Level A, Type II, Method I for boxed tractors.

3.4.2.5 Winterization components. Winterization components shall be preserved as specified in 3.4.1.7.

3.4.2.6 Enclosed gears and chains. The gear housings and chain housings shall be filled to the operating level with the applicable type and grade of lubricant required for operation. A tag as specified in 3.4.2.2 shall be attached to each housing in a conspicuous location indicating: "This housing is filled with the applicable type and grade of lubricant required for operation."

3.4.2.7 Exposed gears. Unpainted surfaces of exposed gears shall be preserved as specified in 3.4.1.9.1 for Level A.

3.4.2.8 Exposed drive chains. **Exposed drive chains shall be coated with Type P-3 or P-9 preservative.**

3.4.2.9 Clutches.

3.4.2.9.1 Disc-type. Spring-loaded-type clutches shall have the clutch pedal secured in a partially disengaged position. Snap-over-center- and toggle-in-type clutches shall be completely disengaged.

3.4.2.9.2 Oil-type and hydraulic boosters. Oil-type flywheel and steering clutches and hydraulic boosters shall be preserved as specified in 3.4.1.11.5, 3.4.1.11.6, 3.4.1.11.7.2, and 3.4.1.11.7.3 as specified for Level A, except tags shall be as specified in 3.4.2.2.

3.4.2.10 Hydraulic systems (including hydraulic brakes). Hydraulic systems (including hydraulic brakes) shall be preserved and tagged as specified in 3.4.1.12.2, 3.4.1.12.4 and 3.4.1.13, except tags shall be as specified in 3.4.2.2.

3.4.2.11 Airbrake systems. Airbrake systems shall be preserved and tagged as specified in 3.4.1.12.3 and 3.4.1.12.4, except tags shall be as specified in 3.4.2.2.

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3.4.2.12 Automatic transmissions, fluid couplings and torque converter housings. Automatic transmissions, fluid couplings and torque converter housings, (except diesel-fuel-operated torque converters) shall be filled to the operating level with the applicable type and grade of fluid or lubricant required for operation. A tag as specified in 3.4.2.2 shall be attached to each control indicating: "The housing is filled with the applicable type and grade of fluid or lubricant required for operation." Diesel-fuel-operated torque converters shall be preserved and tagged as specified in 3.4.1.14.2.2 for Level A, except tags shall be as specified herein.

3.4.2.13 Rubber tires. Rubber tires shall be inflated as specified in 3.3.1.16 for Level A.

3.4.2.14 Instrument panels. For Type II unboxed (mobile) tractors, instruments and instrument panels shall be protected as specified in 3.4.1.18 for Level A.

3.4.2.15 Lamps, reflectors and rear-vision mirrors. Lamps, reflectors and rear-vision mirrors, if removed, shall be preserved as specified in 3.4.1.19 for Level A.

3.4.2.16 Drivebelts. Drivebelts shall be released from tension. A tag shall be attached in a conspicuous location within the operator's compartment indicating: "Tension has been released on all drivebelts. Adjust tension prior to operation." Tags shall be as specified in 3.4.2.2.

3.4.2.17 Seatbacks, armrests and cushions. For Type II unboxed (mobile) tractors not equipped with fully enclosed cabs, the seatbacks, armrests and cushions shall be protected as specified in 3.4.1.22 for Level A.

3.4.2.18 Windshield wiper arms and blades. For Type II unboxed (mobile) tractors, windshield wiper arms and blades shall be removed and packaged in the supplier's commercial box.

3.4.2.19 Locks and keys. Switch keys, cab door keys and padlocks with keys shall be placed in the toolbox. A tag shall be attached in a conspicuous location in the operator's compartment indicating: "Keys for this tractor are placed in the toolbox." Tags shall be as specified in 3.4.2.2.

3.4.2.20 Fire extinguishers. Fire extinguishers shall be secured in their mounting brackets.

3.4.2.21 Maintenance tools, technical publications and repair parts. Maintenance tools, technical publications and repair parts shall be preserved as specified in 3.4.1.30, 3.4.1.31 and 3.4.1.32 for Level A.

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3.4.2.22 Consolidation. Consolidation of components removed from tractors, and attachments shall be as specified in 3.4.1.33.1 and 3.4.1.33.2 for Level A, except in all cases, boxes shall be Domestic Type and strapping of boxes for mobile shipment shall be either Finish A or B.

3.4.3 Commercial. Tractors, components, repair parts, tools, and publications shall be preserved in accordance with FED. STD. No. 356. The crankcases shall contain the required amount of lubricating oil, and the cooling systems shall be filled with a fresh, clean solution of 50 **percent water and 50 percent antifreeze conforming to MIL-A-46153.**

3.5 Packing. Packing shall be Level **A, B, or commercial as specified** (see 6.2).

3.5.1 Level A.

3.5.1.1 Type I boxed tractors. The tractor, and attachments that are practical to pack with the tractor, consolidated items and all other components, comprising each complete tractor shall be packed together in a crate conforming to MIL-C-104, Class I, Style a or b, as applicable. The tractor and components, including wheels if removed, attachments, and consolidated containers shall be securely blocked, braced and anchored within the crate in accordance with MIL-STD-1186. Attachments such as dozers, buckets, booms, or cranes, which are impractical to pack in the crate with the tractor, shall be packed for shipment as separate items as specified in 3.5.1.3. The attachments packed as separate items shall be shipped at the same time as the tractor.

3.5.1.2 Type II unboxed (mobile) tractors. The tractor shall be prepared for unboxed shipment. Consolidated containers and other components shall be positioned on the tractor in any available space that will not increase cubage, damage the tractor, or interfere with towing, lifting with slings, or steering the tractor while being towed. Containers and components shall be secured to the tractor with strapping conforming to QQ-S-781, Class 1, Type I or IV, Finish B, size as applicable, or secured by bolting. When specified (see 6.2), attachments shall be positioned in their normal operating or transport position, whichever is the most convenient for handling and shipment, and shall be secured in position with flat steel strapping as specified herein, or with wire rope. In addition, the attachments shall be blocked and braced as required to prevent movement or damage. Attachments such as dozers, buckets and booms that cannot be relocated on and secured to the tractor shall be packed as specified in 3.5.1.3. The attachments packed as separate items shall be shipped at the same time as the tractor.

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3.5.1.2.1 Cabs. Cab windows and doors shall be closed. When specified (see 6.2), windshields and window glass shall be protected from breakage by covering each glass surface with a fitted piece of hardboard, or plywood 1/4-inch thick, or with a sheet metal panel secured in place. There shall be not less than a 1-inch clearance between the panels and the glass. Small openings into the cabs shall be sealed with tape conforming to PPP-T-60, Type IV, Class 1.

3.5.1.3 Attachments. Unless otherwise specified, attachments such as dozers, buckets and booms shall be prepared for shipment unboxed. To the maximum extent possible, related parts and consolidated components shall be secured to the attachments with steel strapping as specified herein. Boxed and unboxed attachments and consolidated items that cannot be secured to the attachment or tractor shall be shipped separately from but at the same time as the tractor. When specified (see 6.2), attachments such as dozers, tree-dozers, bull clams, loaders, pipelayers, mowers, sweepers, cranes, component parts, and consolidated items comprising complete attachments shall be packed in crates conforming to PPP-C-650, Style A, Type V, Class 1, or MIL-C-3774, Type I. The contents of the crates shall be blocked, braced and anchored in accordance with MIL-STD-1186. Strapping used for securing of the crate contents shall conform to QQ-S-781, Class 1, Type I or IV, Finish B, size as applicable.

3.5.2 Level B.

3.5.2.1 Type I boxed tractors. The tractor and its components shall be packed as specified in 3.5.1.1 for Level A. As an alternate, crates shall conform to MIL-C-3774, Type I.

3.5.2.2 Type II unboxed (mobile) tractors. The tractor and components shall be packed as specified in 3.5.1.2. Strapping shall be either Finish A or B.

3.5.3 Commercial. The tractors, components, attachments, repair parts, tools, and technical publications shall be prepared for unboxed (mobile) shipment in compliance with FED. STD. No. 356.

3.6 Marking.

3.6.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.

3.6.2 Commercial. Marking for commercial packaging shall be in accordance with FED. STD. No. 356.

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3.7 Depreservation guides.

3.7.1 Military packaging. Two depreservation guides shall be prepared. One guide shall be placed in a waterproof envelope marked "Depreservation Guide" and secured in a conspicuous location on the steering wheel or steering controls of the tractor. The other guide shall be placed in the package of technical manuals and the package marked to indicate the depreservation guide is inside. Unless otherwise specified (see 6.2), DA Form 2258 "Depreservation Guide for Vehicles and Equipment" shall be used.

3.7.2 Commercial packaging. Two depreservation guides as specified in 3.7.1 or two sets of commercial instructions for readying the vehicles for operation, shall be furnished. The guides or instructions shall be located as specified in 3.7.1.

3.8 Workmanship. All operations and processes involved in accomplishing the cleaning, preserving, packaging, packing, and marking requirements specified herein shall be in accordance with the highest grade practices associated with this type of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that supplies and materials used are manufactured, examined and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction pack inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

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4.3 Preproduction pack inspection.

4.3.1 Examination. The preproduction pack shall be examined for the defects specified in 4.4.4. Examination shall be in three stages as specified in 4.4.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Test. The preproduction pack of boxed tractors and of boxed attachments for Level A or B shall be tested as follows: The boxed tractor and the boxed attachments shall be subjected to the railroad car test as specified for guided-impact tests in accordance with MIL-STD-1186, Appendix A. The car shall strike a string of five empty cars, with draft gear extended and the brakes set, at a speed of not less than 10 miles per hour. At the conclusion of the preproduction pack test, the pack shall be examined. Any shifting of contents, loosening or breaking of holddowns, ties, stays, blocking, or bracing, or any visible damage to the tractors or attachments shall constitute failure of the preproduction pack and shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Inspection stages.

(a) The first stage shall include inspection of procedures, containers, materials, methods, marking, and placement of components in consolidated containers.
(b) The second stage shall include inspection of the marking and closure of the consolidated containers and the securing of the containers and components to the shipping crates and unboxed (mobile) tractors.
(c) The third stage shall include inspection of materials and marking of uncrated items and crated items after closure.

4.4.2 Unit of product. For the purpose of inspection, a completed pack prepared for any applicable inspection stage shall be considered a unit of product.

4.4.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.4 Examination. Samples selected in accordance with 4.4.3 shall be examined for the defects marked "X" for the applicable level in Table I. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

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Table I. Classification of Defects

Defect	Level A	Level B	Level C	Requirement Paragraph
101. Materials, methods and containers not as specified for preservation, packing, and marking. Each incorrect material, method or container shall constitute one defect.	X	X	X	3.1, 3.4, 3.5, 3.6
102. Damaged portions of painted surfaces not repainted as specified.	X	-	-	3.3
103. Disassembly not as specified.	X	X	X	3.4.1.1, 3.4.2.1, 3.4.3.3
104. Matchmarking not as specified.	X	X	X	3.4.1.2, 3.4.2.1, 3.4.3.3
105. Tags used for matchmarking not as specified.	X	X	-	3.4.1.3, 3.4.2.2
106. Printing on tags not waterproof.	X	-	-	3.4.1.3
107. Unprotected metal surfaces not coated with preservatives as specified.	X	X	-	3.4.1.5, 3.4.2.3
108. Preserved machined surfaces not wrapped or covered with barrier material, and the wraps or covers not secured with tape as specified.	X	X	-	3.4.1.5, 3.4.2.3
109. Engines and engine components not preserved as specified.	X	X	X	3.4.1.6, 3.4.2.4, 3.4.3.2, (when specified)
110. Pumps, burners and torches not preserved as specified.	X	X	-	3.4.1.7.1, 3.4.2.5
111. Blowtorches and blowtorch type burners not placed in fiberboard boxes as specified.	X	X	-	3.4.1.7.1, 3.4.2.5
112. Heater fuel tanks not preserved as specified.	X	X	-	3.4.1.7.2, 3.4.2.5
113. Electric motors not preserved in accordance with referenced document.	X	X	-	3.4.1.7.3, 3.4.2.5
114. Drive shafts and universal joints not coated with preservative as specified and dust boots not reinstalled as specified.	X	-	-	3.4.1.8
115. Exposed gears not coated with preservative or with primer as specified.	X	X	-	3.4.1.9.1, 3.4.2.7

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Table I. Classification of Defects (Cont'd)

Defect	Level A	Level B	Level C	Requirement Paragraph
116. Enclosed gear housing not filled with preservative oils or with operational lubricant as specified and the housings not tagged as specified.	X	X	-	3.4.1.9.2, 3.4.2.6
117. Exposed drive chains not coated with preservative as specified.	X	X	-	3.4.1.10.1, 3.4.2.8
118. Enclosed chain housings not filled with preservative oil or not filled with operational lubricant as applicable, and the housings not tagged as specified.	X	X	-	3.4.1.10.2, 3.4.2.6
119. Dry-type clutch components not coated as specified.	X	-	-	3.4.1.11.1, 3.4.1.11.2, 3.4.1.11.3, 3.4.1.11.4
120. Disc-type and cone-type clutches not partially or fully disengaged as specified.	X	X	-	3.4.1.11.1, 3.4.1.11.3, 3.4.2.9.1
121. Flywheel clutches not preserved and tagged as specified.	X	X	-	3.4.1.11.5, 3.4.1.11.6, 3.4.2.9.2
122. Steering clutches not preserved and not tagged as specified.	X	X	-	3.4.1.11.7, 3.4.2.9.2
123. Hydraulic boosters not preserved and not tagged as specified.	X	X	-	3.4.1.11.7.3, 3.4.2.9.2
124. Clutch housing drainplugs for dry-type clutches for Type I boxed tractors not removed, not preserved, and not tagged as specified.	X	-	-	3.4.1.11.8
125. Openings into clutch housings for Type II unboxed (mobile) tractors not covered and not tagged as specified.	X	-	-	3.4.1.11.8
126. Exterior surfaces of brakes not coated with preservative as specified.	X	-	-	3.4.1.12.1
127. Surfaces of brake components enclosed within brakedrums and the braking surfaces of brakedrums not coated with primer when specified and not tagged as specified.	X	-	-	3.4.1.12.1

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Table I. Classification of Defects (Cont'd)

Defect	Level A	Level B	Level C	Requirement Paragraph
128. Hydraulic brake systems operating on nonpetroleum base fluid not filled with preservative fluid and not tagged as specified.	X	X	-	3.4.1.12.2.1, 3.4.2.9.2
129. Hydraulic brake systems operating on petroleum base fluid not filled to the operating level with the fluid required for operation and the system not tagged as specified.	X	X	-	3.4.1.12.2.2, 3.4.2.9.2
130. Air-actuated brake system not preserved and not tagged as specified.	X	X	-	3.4.1.12.3, 3.4.2.11
131. Air hydraulic brakes not preserved and not tagged as specified.	X	X	-	3.4.1.12.4, 3.4.2.11
132. Automatic transmissions, fluid couplings, and torque converters not preserved and tagged as specified or not filled with applicable type lubricant and tagged as specified.	X	X	-	3.4.1.14, 3.4.2.12
133. Wheels, axles and bearings not preserved as specified.	X	-	-	3.4.1.15
134. Rubber tires not inflated to pressure value specified.	X	X	-	3.4.1.16, 3.4.2.1
135. Crawler components not coated with preservative as specified.	X	-	-	3.4.1.17
136. Instrument panels for Type II unboxed (mobile) tractors not protected as specified.	X	X	-	3.4.1.18, 3.4.2.14
137. Trouble lights, lamps, reflectors, and rear-vision mirrors not boxed and, for Type II unboxed (mobile) tractors, not protected as specified.	X	X	-	3.4.1.19, 3.4.2.15
138. Fiberboard boxes not closed as specified.	X	X	-	3.4.1.7.1, 3.4.1.19, 3.4.1.29, 3.4.2.5, 3.4.2.15
139. Gearshifts and control levers not preserved as specified and, for Type II unboxed (mobile) tractors, not equipped with a full cab, composition and rubber handgrips not preserved or protected as specified.	X	-	-	3.4.1.20

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Table I. Classification of Defects (Cont'd)

	Defect	Level A	Level B	Level C	Requirement Paragraph
140.	Drivebelts not removed or released from tension and not tagged as specified.	X	X	-	3.4.1.21, 3.4.2.16
141.	Pulley grooves not coated with primer as specified.	X	-	-	3.4.1.21
142.	Seatbacks, armrests and cushions for Type II unboxed (mobile) tractors not equipped with closed cabs not wrapped or protected as specified.	X	X	-	3.4.1.22, 3.4.2.17
143.	Windshield wiper arms and blades for Type II unboxed (mobile) tractors not removed and preserved as specified and wiper shafts not coated with preservative as specified.	X	X	-	3.4.1.23, 3.4.2.18
144.	Interior of locks not coated with powdered graphite and keys not preserved as specified.	X	-	-	3.4.1.24
145.	Openings into exterior locks for Type II unboxed (mobile) tractors not sealed with tape as specified.	X	-	-	3.4.1.24
146.	Keys and padlocks not placed in toolbox as specified.	X	X	-	3.4.1.24, 3.4.2.19
147.	Tag not attached in operator's compartment indicating keys and padlocks are in toolbox.	X	X	-	3.4.1.24, 3.4.2.19
148.	Hinges and fasteners not coated with preservative as specified.	X	-	-	3.4.1.25
149.	Engine hoods not closed and secured as specified.	X	-	-	3.4.1.26
150.	Vents, louvers and other openings into engine hoods not covered for Type II unboxed (mobile) tractors.	X	-	-	3.4.1.27
151.	Radiator fronts for Type II unboxed (mobile) tractors not covered when specified.	X	-	-	3.4.1.27
152.	Attachments not preserved, and positioned as specified.	X	-	-	3.4.1.28.1 3.4.1.28.2, 3.4.1.28.4, 3.4.1.28.5

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Table I. Classification of Defects (Cont'd)

	Defect	Level A	Level B	Level C	Requirement Paragraph
153.	Wire rope (except galvanized) not coated with preservative as specified and not rewound on drum as specified.	X	-	-	3.4.1.28.3
154.	Wire rope for Type II unboxed (mobile) tractors not covered as specified.	X	-	-	3.4.1.28.3
155.	Unpainted exterior surfaces of fire extinguishers not coated with preservative as specified.	X	-	-	3.4.1.29
156.	Fire extinguishers not secured in mounting brackets or placed in a box as specified	X	X	-	3.4.1.29, 3.4.2.20
157.	Maintenance tools not preserved in accordance with referenced document.	X	X	X	3.4.1.30, 3.4.2.21, 3.4.3.1
158.	Technical publications not preserved as specified.	X	X	X	3.4.1.31, 3.4.2.21, 3.4.3.1
159.	Repair parts not preserved in accordance with the referenced document.	X	X	X	3.4.1.32, 3.4.2.21, 3.4.3.1
160.	Consolidation not as specified.	X	X	X	3.4.1.33, 3.4.2.22, 3.4.3.1
161.	Toolbox lids not secured to prevent pilferage.	X	X	X	3.4.1.33, 3.4.2.22, 3.4.3.1
162.	Packing not as specified.	X	X	X	3.5
163.	Blocking, bracing and anchoring not as specified.	X	X	-	3.5.1.1, 3.5.1.3, 3.5.2.1
164.	Positioning of consolidated components on tractors prepared for unboxed (mobile) shipment not as specified.	X	X	-	3.5.1.2, 3.5.2.21
165.	Attachments not positioned as specified, when required, for Type II, unboxed (mobile) tractors.	X	X	-	3.5.1.2
166.	Windshields and window glass not protected as specified, when required, and cab openings not sealed as specified.	X	X	-	3.5.1.2.1, 3.5.2.2

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Table I. Classification of Defects (Cont'd)

Defect	Level A	Level B	Level C	Requirement Paragraph
167. Strapping not as specified.	X	X	-	3.4.1.33.2, 3.4.1.33.3, 3.4.2.22, 3.4.3.1, 3.5.1.2, 3.5.1.3, 3.5.2.2
168. Marking missing, illegible, incomplete, or incorrect.	X	X	X	3.6
169. Depreservation guides not prepared and stowed as specified.	X	X	X	3.7

5. PREPARATION FOR DELIVERY

(This section not applicable to this specification).

6. NOTES

6.1 Intended use. It is intended that this specification cover preservation, packing, and marking of tractors and their attachments, and shall be used for reference in Section 5 of equipment specifications, direct references in contract or orders, or in the preparation of packaging data sheets.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of shipment required (see 1.2)
(c) Time frame required for submission of the preproduction pack (see 3.2).
(d) Preservation and packing required (see 3.4 and 3.5). Level B preservation and packing is intended to provide limited but economical protection and should be specified only when it is determined the tractors and attachments will be shipped in covered carriers and held in covered storage.
(e) When disassembly of the clutch is required for preservation (see 3.4.1.11.1).
(f) When brake components enclosed within brakedrums and the braking surfaces of brakedrums are to be preserved (see 3.4.1.12.1).

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(g) When radiator fronts are to be protected (see 3.4.1.27).
(h) When repair parts should be preserved in accordance with MIL-R-196 (see 3.4.1.32 and 3.4.2.21).
(i) When other than Finish B strapping is required (see 3.4.1.33.2 and 3.4.1.33.3).
(j) Requirements for Level B preservation if other than as specified (see 3.4.2 and 6.6).
(k) When alternate commercial preservation is required (see 3.4.3).
(l) When attachments are to be positioned in normal or transport position (see 3.5.1.2).
(m) When windshields and window glass are to be protected (see 3.5.1.2.1).
(n) When attachments are to be prepared for boxed shipment (see 3.5.1.3).
(o) When other than DA Form 2258 is to be used (see 3.7).

6.3 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the supplier of his obligation to preserve, pack, and mark the tractors and attachments in accordance with this specification.

6.4 The requirements of 3.3 are intended to apply only in those cases where the painting requirements of the equipment specification would not otherwise be applicable, such as on separate packaging contracts or in depot operations.

6.5 The percentage of fuel oil contained in preservative oil which has been used to preserve burner fuel tanks (see 3.4.1.7.2 and 3.4.2.5) may be determined within practical limits by use of American Petroleum Institute hydrometers and control specimens of known dilution.

6.6 Level B preservation (see 3.4.2) has been developed for use by the Army and Air Force for the specific conditions specified in 6.2. For Navy procurements, details of Level B preservation, when required, should be developed at the time of procurement when all conditions are known.

6.7 Depreservation guide. The contracting officer should arrange to furnish the necessary copies of DA Form 2258 when requested by the supplier (see 3.7).

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6.8 Environmental pollution preventive measures are contained in the material specifications indicated herein. Refer to material specifications (or preparing activity) for recommended disposability methods.

Custodians:

Army - ME
Navy - YD
Air Force - 84

Preparing activity:

Army - ME

Project No. PACK-0522

Review activities:

Army - SM, AT
Navy - MC

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	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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